

This policy cross-reference matrix maps mandatory governance gates and required compliance artifacts across all six lifecycle phases—Discover, Define, Design, Develop, Delivery, and Run—for pharmaceutical drug R&D through commercial manufacturing portfolios.

1. R&D Portfolio

Phase	Required Artifacts	Mandatory Reviews & Gates
Discover	Target Product Profile (TPP) Draft, Target Discovery Feasibility Study	Gate 0: Candidate Selection Approval
Define	Critical Quality Attributes (CQAs) Matrix, Analytical Method Specifications	SRR: Discovery Quality & Target Review
Design	Formulation & Process Design Space, High-Throughput Screening Models	PDR/CDR: Lead Candidate Freeze
Develop	Pre-Clinical Safety Data, Assay Validation Reports	TRR: IND/CTA Submission Readiness Gate
Delivery	Investigational New Drug (IND) Dossier, Regulatory Submission Pack	Regulatory Filing Approval Sign-off
Run	Patent Maintenance Logs, Post-Discovery Line Extension Studies	Annual Portfolio Strategy Review

2. Clinical Trials Portfolio

Phase	Required Artifacts	Mandatory Reviews & Gates
Discover	Clinical Strategy Feasibility Study, Patient Population Analytics	Gate 0: Clinical Investment Gate
Define	Master Clinical Protocol, eCOA Strategy, IRB/Ethics Requirements	SRR: Clinical Protocol Review
Design	eCRF Design Specifications, Decentralized Clinical Trial (DCT) Architecture	PDR/CDR: Protocol Architecture Freeze
Develop	EDC Validation Reports, Clinical Trial Material (CTM) Packaging Workflows	TRR: First-Patient-In (FPI) Gate

Phase	Required Artifacts	Mandatory Reviews & Gates
Delivery	Clinical Study Reports (CSR), Blinded Data Integrity Audit Sign-offs	Trial Phase Completion Gate
Run	Post-Marketing Trial Dashboards, Real-World Evidence (RWE) Analytics	Annual Clinical Governance Audit

3. Tech Transfer Portfolio

Phase	Required Artifacts	Mandatory Reviews & Gates
Discover	Receiving Site Capacity Assessment, Scale-Up Feasibility Study	Gate 0: Tech Transfer Approval
Define	Tech Transfer Plan (TTP), Process Gap Analysis, Equipment Mapping	SRR: Scale-Up Requirements Review
Design	Process Analytical Technology (PAT) Specs, Scale-Up Digital Twin Models	PDR/CDR: Process Transfer Freeze
Develop	Engineering Batch Protocols, Scale-Down Model Verification Data	TRR: Pilot-Scale Engineering Gate
Delivery	Process Performance Qualification (PPQ) Reports, Site Dossier Sign-offs	Commercial Site Readiness Gate
Run	Continued Process Verification (CPV) Logs, Yield Drift Analytics	Annual Tech Transfer Review

4. Shop Floor & MES Portfolio

Phase	Required Artifacts	Mandatory Reviews & Gates
Discover	Digital Plant Vision, Automation & Robotics Feasibility Study	Gate 0: Modernization Approval
Define	Electronic Batch Record (eBR) Functional Specs, Equipment IoT Specs	SRR: MES Architecture Review
Design	Master Recipe Specs, Weigh & Dispense Digital Twin Models	PDR/CDR: eBR Design Freeze
Develop	Core MES Workflows, PLC/SCADA Integration Protocols	TRR: Pilot Line Readiness Review
Delivery	CSV Computerized Systems Qualification Pack, FAT Reports, Operator Kits	Gate 3: Plant Go-Live Gate
Run	Real-Time OEE Reports, Batch Release Lead-Time Logs	Monthly Operations Review

5. MFG Operations Portfolio

Phase	Required Artifacts	Mandatory Reviews & Gates
Discover	Facility Capacity Strategy, Continuous Manufacturing Feasibility	Gate 0: Operations Strategy Approval
Define	Master Production Schedule (MPS) ConOps, Environmental Monitoring Specs	SRR: Manufacturing Requirements
Design	Cleanroom Workflow Models, Automated Unit Operation Schemas	PDR/CDR: Layout & Workflow Freeze
Develop	Master Production Records (MPR), Cleaning Validation Protocols	TRR: Facility Qualification Review
Delivery	Commercial Batch Release Authorization, Regulatory License Packs	Commercial Launch Go-Live Gate
Run	Batch Yield Analytics, Right-First-Time (RFT) Tracking Logs	Monthly Plant Governance Review

6. Supply Chain Operations Portfolio

Phase	Required Artifacts	Mandatory Reviews & Gates
Discover	Cold-Chain Maturity Assessment, Distribution Network Strategy	Gate 0: Supply Chain Strategy Gate
Define	Control Tower Specs, DSCSA Serialization Standards	SRR: Requirements Review
Design	Cold-Chain Monitoring Architecture, ERP-WMS Integration Specs	PDR/CDR: Logistics Design Review

Phase	Required Artifacts	Mandatory Reviews & Gates
Develop	Automated Track-and-Trace Workflows, Temperature Telemetry Pipeline	TRR: Logistics Integration Test
Delivery	Serialized Product Tracking Sign-offs, Distribution Playbook	Operations Approval Sign-off
Run	Lead-Time Dashboards, Real-Time Temperature Excursion Logs	Quarterly Supply Chain Review

7. Vendor Management Portfolio

Phase	Required Artifacts	Mandatory Reviews & Gates
Discover	Raw Material Sourcing Strategy, Vendor Risk Matrix	Gate 0: Sourcing Program Gate
Define	Supplier Quality Agreement Templates, EDI Integration Standards	SRR: Vendor Requirements Review
Design	Vendor Portal UX, Automated Audit Workflow Schemas	PDR/CDR: Supplier Portal Freeze

Phase	Required Artifacts	Mandatory Reviews & Gates
Develop	Supplier API Connectors, Automated Certificate of Analysis (CoA) Ingestion	TRR: Vendor Integration Test
Delivery	Qualified Supplier List (QSL) Sign-offs, Vendor Onboarding Playbook	Supplier Qualification Gate
Run	Automated Vendor Risk Scoring Logs, Scorecard Dashboards	Quarterly Vendor Governance Review

8. Predictive Maintenance Portfolio

Phase	Required Artifacts	Mandatory Reviews & Gates
Discover	Facility Sensor Feasibility Study, Asset Maintenance Strategy	Gate 0: Predictive Asset Gate
Define	Bioreactor & HVAC Telemetry ConOps, Asset Data Strategy	SRR: Asset Requirements Review
Design	Predictive Machine Failure ML Models, Sensor Schemas	PDR/CDR: Algorithm Design Freeze
Develop	ML Telemetry Pipeline Build, Automated Maintenance Workflows	TRR: Sensor Validation Review

Phase	Required Artifacts	Mandatory Reviews & Gates
Delivery	Equipment Calibration Sign-offs, Digital Maintenance Playbook	Asset Readiness Gate
Run	Real-Time Anomaly Alerts, Equipment Dashboards, ML Retraining Logs	Reliability Review Board

9. On-Demand Repair Portfolio

Phase	Required Artifacts	Mandatory Reviews & Gates
Discover	Downtime Benchmarking Study, Rapid Repair Strategy	Gate 0: Rapid Service Gate
Define	Rapid Engineering Change ConOps, Emergency Spare Part Specs	SRR: Repair Requirements Review
Design	Approved 3D-Printable Spare Part CAD Specs, Rapid Workflows	PDR/CDR: Repair Standards Review
Develop	Mobile Maintenance App, Automated Work Order Generation Pipelines	TRR: Field Simulation Test

Phase	Required Artifacts	Mandatory Reviews & Gates
Delivery	Command Center Playbook, Emergency Maintenance Tooling Packs	Service Network Go-Live Gate
Run	Mean Time to Repair (MTTR) Logs, Emergency Dispatch Metrics	Monthly Service Governance

10. Quality Intelligence Portfolio (Inspection → AI Prediction & Prevention)

Phase	Required Artifacts	Mandatory Reviews & Gates
Discover	Quality 4.0 Vision, AI Vision & Acoustic Sensing Feasibility Study	Gate 0: Quality Transformation Gate
Define	Digital QMS Architecture, Out-of-Specification (OOS) AI Specs	SRR: Requirements Review
Design	AI Defect Models, Automated Visual Particle Inspection Specs	PDR/CDR: Quality Model Freeze
Develop	Vision Training Datasets, Automated CAPA Pipeline Code	TRR: Model Validation Review

Phase	Required Artifacts	Mandatory Reviews & Gates
Delivery	Automated CoA Generation Sign-offs, Regulatory Audit Dossiers	Health Authority Quality Sign-off
Run	Real-Time Defect Prediction Dashboards, Root Cause Logs, AI Drift Reports	Quality Board cGMP Review

11. Customer Support Portfolio

Phase	Required Artifacts	Mandatory Reviews & Gates
Discover	Patient/Provider Experience Benchmarks, MedInfo Strategy	Gate 0: Experience Investment Gate
Define	Medical Inquiry Portal Specs, Adverse Event (AE) Intake Rules	SRR: Portal Requirements Review
Design	MedInfo Portal Architecture, Automated AE Reporting Workflows	PDR/CDR: Support Systems Freeze
Develop	Omnichannel Customer Portal, 24/7 Global Desk Integration	TRR: Desk Simulation Test

Phase	Required Artifacts	Mandatory Reviews & Gates
Delivery	Commercial Launch Playbook, Patient Assistance Program Integration	Final Commercial Acceptance
Run	Net Promoter Score (NPS) Dashboards, Pharmacovigilance (PV) Logs	Annual Operations Review